Department of Electrical Engineering, University of Cape Town
EEE4022S Final Year Project 2026

Embedded Telemetry with Road-Surface Characterisation and Camera-Based Hazard Detection for the TUKZIE Electric Trike

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1 Problem and aim

- Electric cargo trikes such as the TUKZIE carry last-mile freight cheaply, but their ride and the roads they use go unmeasured.
- **Aim:** a low-cost unit that measures the ride at both ends, warns the driver of hazards and keeps its data through coverage gaps.



Road test, 27 Oct

2 System on the trike, 27 Oct

Front: Raspberry Pi 5

- Front camera, detector
- Dashboard, 10.1 in screen
- Front node: ESP32-S3, IMU at 200 Hz
- Recorder; phone GNSS

Ethernet

Rear: Raspberry Pi 4

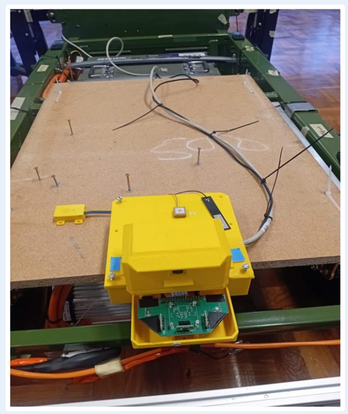
- Rear camera, detector
- Rear node: ESP32-S3, IMU at 200 Hz
- Telemetry unit: ESP32-S3 with battery, GNSS and LTE



Front computer



Front camera



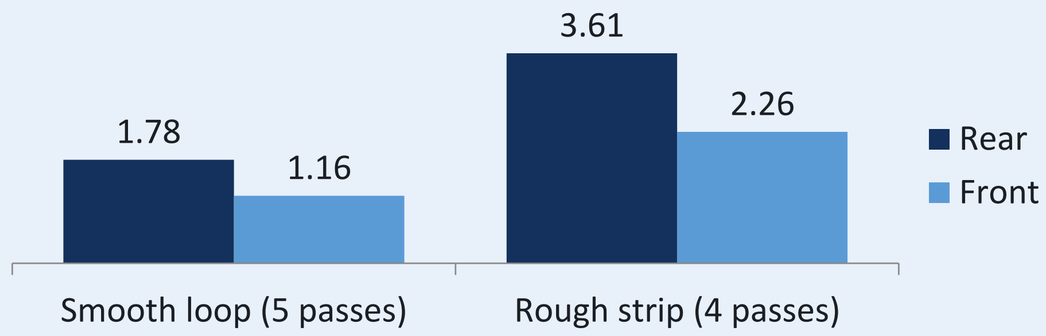
Rear unit

3 Method

- **Requirements first:** thirteen requirements, each with a criterion fixed before its test; bench tests, then the vehicle.
- **Road, 27 Oct:** five passes of a smooth loop and four of a rough strip at 10 km/h; phone GNSS as the speed reference.
- **Analysis:** ISO 2631-1 weighted vibration a_w per pass; CV over passes (criterion 10 %); ICC in 8 m cells (0.75); share of the stand-run motor lines in the road spectrum.

4 Key results on the road

Mean weighted vibration per pass, 10 km/h (m/s²)



5.0 % / 5.3 %
CV of a_w over 5 passes, rear / front; criterion 10 %: met

about 2 ×
rough strip against the smooth loop, at both ends

ICC 0.16 vs 0.75
not met: repeats over whole passes, not place by place

0.9 to 3.1 %
of road vibration energy in the motor lines, powered passes

0 gaps
front IMU at 200 Hz; rear 0.08 % of samples lost

99.9 %
of records over the wired link to the Raspberry Pi 4

153 records
over LTE, 116 replayed after the drop; median 0.44 s (n = 9)

23.3 / 12.1 fps
detection, front / rear; median 68 / 103 ms

5 Limitations

- One vehicle, two short low-speed sessions; no reference accelerometer, so repeatability, not accuracy.
- LTE did not reconnect after 11:51; the vehicle's GNSS lay 8 to 25 m from the phone's.
- Warnings below ISO 15008 legibility; the driver's glance time was not measured.

6 Conclusion

- The unit ran on the trike; its ride reading repeats over whole passes, separates two surfaces and is dominated by the road.
- Next: own supply, larger warnings, glance time.

Videos of the system on the road
tuksense.site